MATH 107: Introduction to Linear Algebra

Fall 2026 · MoWeFr 10:00–10:50 a.m. · FR 201

Quick Start

- Start with the course schedule for the recommended path through the material.
- See the syllabus for policies and grading.
- Readings live under lectures.
- Assessments lists labs, quiz prep, exams, and the project.
- Software and references are on resources.
- Canvas is where you submit work, take graded quizzes, see grades, and get feedback.

Instructor

Dr. Rosanna Overholser
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Office: BSS 334
Office hours: Thursdays 11:00–11:50 a.m. in BSS 334; Wednesdays 12:30–2:30 p.m. in the Learning Center (first floor of the library); or by appointment.

Welcome

This website contains materials for **MATH 107: Introduction to Linear Algebra** (3 units, GE Area 2) at Cal Poly Humboldt. Class meets face to face Monday, Wednesday, and Friday, 10:00–10:50 a.m., in FR 201.

Catalog description: Euclidean spaces, matrices and matrix arithmetic, solving systems of equations, and eigenvalues and eigenvectors. Emphasis on applications and use of computation.

This site follows three units: representing and comparing things with vectors; transforming, solving, simulating, and computing with matrices; then explaining dynamics, fitting data, and exploiting structure (eigenvectors, least squares, SVD). Daily learning objectives are on Lectures. Class outlines for later meetings will be posted before those days.

Where you are

Most of what you read lives on this website: lecture outlines, labs, quiz preparation, exam reviews, and project instructions. Canvas is where you submit work, take graded quizzes, see grades, get feedback, and chat with classmates.

What you'll learn

By the end of the course you should be able to:

1. Interpret matrices, vectors, and complex numbers, both algebraically and geometrically.
2. Solve systems of linear equations using row-reduction techniques by hand and with software.
3. Find eigenvalues and eigenvectors of a matrix by hand for a 2×2 matrix and with software for general cases.
4. Understand and apply eigenvalues and eigenvectors to solve problems.
5. Recognize special types of matrices and their properties, such as triangular, symmetric, and positive definite matrices.
6. Communicate in the language of linear algebra, including linear combination, subspace, span, independence, basis, and dimension.
7. Write or adapt programs to implement linear algebra concepts, such as matrix arithmetic, vectors, and plotting.
8. Apply linear algebra to real-world problems.
9. Learn independently as well as collaboratively. Develop study techniques that include pre- and post-reading of class materials and be able to critically analyze and implement solutions in a team setting.

Daily objectives for each meeting are on the lectures page.